IMPLEMENTING BOUNDED COMMUTING PROJECTIONS IN FEEC

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Abstract. Bounded commuting projections are essential for the analysis of finite element exterior calculus (FEEC) applied to various PDEs. The canonical projections are unfortunately not bounded in the L^2 norm. Thus, researchers have worked on constructing other projections that are both commuting and bounded. In this report, we review a projections developed by [Ern et al., FOCM, 2026] and discuss some aspects of generating an NGSolve code the implements their projections. Using tetrahedral meshes, we work with the de Rham sequence and define projections onto the FEEC spaces. The key idea is to build weight functions that live on FEEC spaces themselves. To do this, we have to solve them in a two-step process. In the second step, there is a small linear system we have to solve for each geometric object (e.g., vertices, edges, faces, tetrahedra of the mesh).

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1. Introduction

A corner stone in the stability of the finite element exterior calculus (FEEC) methods applied to several PDEs (e.g. Hodge Laplacian) is the existence of bounded commuting projections [3, 2, 5]. In fact, existence of bounded commuting projections leads discrete Poincare inequalities [3, Theorem 5.11] and the discrete Poincare inequalities lead to stability [3, Theorem 7.3]. In [3] they use the Christiansen-Winther projection [7]. Although, this projection is bounded and commuting it is not locally defined which might be desirable.

The first local, bounded and commuting projections were developed by Falk and Winther [9] which was bounded in the graph norm. More recently, Arnold and Guzmán [1] constructed a projection that is local and bounded in L^2. The projection in [1] used a local, continuous operator to construct it. Using similar ideas Ern et al. [8] used only local discrete operators to construct a local, bounded and commuting projections. Hence, one theoretically can compute the projections using FEM software. This is exactly what we aim to do in this project.

In this project, our aim was to use the software package NGSolve [12] to implement the projections define in [8]. The projections are defined via local weight functions (i.e. Z^k below) and the involved part of the code is the constructing of the weights. For every sub-simplex (e.g. vertex, edge, face, tetrahedra) of the mesh there is a corresponding weight function. To solve for such a weight one needs to solve a linear system that involved local FEEC spaces.

In this report, we recall the main ingredients of the projections constructed in [8]. In particular, we recall the conditions that need to be satisfied by the weight functions. We then describe the linear systems we propose to solve to construct the weights. The implementation of projections will appear soon in Github.

Existence of bounded commuting projections mainly a theoretical tool to prove stability of numerical methods [3]. However, such projections have been used for computations. For example, Gallistl et al. [10] use the Falk and Winther projections [9] for multi-scale problems involving $H(\mathrm{curl})$.

The report is structured as follows. We first go over preliminaries which consists of concepts from topology and vector calculus. We then discuss finite element exterior calculus spaces and canonical projections, which motivate the next sectoin: bounded commuting projections. We discuss the projection constructed in [8], whitney forms, projection and commuting properties. Finally, we discuss the construction of the required weights.

2. Preliminaries

Our preliminary section will cover topological concepts, then some concepts from vector calculus and we conclude with a section on function spaces. Topology is required to define our domain and spaces. Vector

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calculus is required to understand the definitions and key properties of our projections. Function spaces allows us to define what these projections are acting on, and what domains these operators work on.

2.1. **Domain and Mesh.**

Definition 2.1 (A n-dimensional simplex)**.** A n-simplex T is determined by $n+1$ vertices $x_0, \dots, x_n$ and is the set:

$$T = \{\sum_{i=0}^{n} t_i x_i : \sum_{i=0}^{n} t_i = 1, t_0, \dots, t_n \geq 0\}.$$

A line segment in 1d, a triangle in 2d, and a tetrahedra in 3d are examples of an n-simplex T.

Definition 2.2 (Simplicial complex)**.** A simplicial complex K in a Euclidean space is a collection K of simplices such that:

(1) If $T \in K$ then all its sub-simplices also belong to K.
(2) The non empty intersection of two simplices $T_1, T_2 \in K$ is a face both of T_1 and T_2.

Definition 2.3 (Oriented simplex σ)**.** Let T be a simplex with vertices $x_0, \dots, x_p$. Then, we let $\sigma = [x_0, \dots, x_p]$ be an orientated simplex. If one permutes the ordering of the vertices by an even permutation we denote this oriented simplex also by σ. If one permutes the ordering of the vertices by an odd permutation then the result says is the same as $-\sigma$.

We consider $\Omega \subset \mathbb{R}^3$ be a bounded polyhedral domain with boundary $\partial\Omega$. Let $\mathcal{T}_h$ be a **tetrahedral mesh** of Ω, with mesh size h. We let $\Delta(\mathcal{T}_h)$ denotes the set of all subsimplices of $\mathcal{T}_h$, and $\Delta_\ell(\mathcal{T}_h)$ denotes the set of ℓ-dimensional subsimplices. That is,

$\Delta_0(\mathcal{T}_h)$ is the set of all vertices,
$\Delta_1(\mathcal{T}_h)$ is the set of of edges,
$\Delta_2(\mathcal{T}_h)$ is the set of all faces,
$\Delta_3(\mathcal{T}_h)$ is the set of tetrahedron.

From now on we use Δ_ℓ to denote $\Delta_\ell(\mathcal{T}_h)$. Note that $\Omega = \bigcup_{K \in \mathcal{T}_h} K$. We assume $\mathcal{T}_h$ induces a simplicial complex. Hence, given two tetrahedra K and T of $\mathcal{T}_h$ we have that either $K \cap T$ does not intersect or it intersects a common vertex, edge or face of both K and T.

2.2. **Differential Operators.**

We first recall some well known concepts from vector calculus. In these definitions we will be working with differentiable functions such as $u : \mathbb{R}^3 \to \mathbb{R}$ and $F : \mathbb{R}^3 \to \mathbb{R}^3$.

∂_i denotes the ith partial derivative, and, for example, the second partial would be

$$\partial_2 u(x) = \lim_{h \to 0} \frac{u(x_1, x_2 + h, x_3) - u(x_1, x_2, x_3)}{h}.$$

The gradient operator maps a scalar function to a vector-valued function and is given by:

$$\text{grad } u = \begin{bmatrix} \partial_1 u \\ \partial_2 u \\ \partial_3 u \end{bmatrix}.$$

If F is smooth vector valued function

$$F = \begin{bmatrix} F_1 \\ F_2 \\ F_3 \end{bmatrix}.$$

Then, we define the divergence operator as

$$\text{div } F = \partial_1 F_1 + \partial_2 F_2 + \partial_3 F_3.$$

It takes a vector valued function to a scalar valued function.

The curl operator takes a vector valued function to a vector valued function.

$$\operatorname{curl} F = \begin{bmatrix} \partial_2 F_3 - \partial_3 F_2 \\ -(\partial_1 F_3 - \partial_3 F_1) \\ \partial_1 F_2 - \partial_2 F_1 \end{bmatrix}.$$

Some useful properties of the operators is the following

$$\operatorname{curl}(\operatorname{grad}(u)) = 0, \ \operatorname{div}(\operatorname{curl}(F)) = 0.$$

We now move on to define the de Rham complex. The **de Rham complex** (with smooth spaces) on Ω is given by:

$$C^\infty(\overline{\Omega}) \xrightarrow{\text{grad}} [C^\infty(\overline{\Omega})]^3 \xrightarrow{\text{curl}} [C^\infty(\overline{\Omega})]^3 \xrightarrow{\text{div}} C^\infty(\overline{\Omega}).$$

Where $C^\infty(\overline{\Omega})$ denotes all the smooth functions on $\overline{\Omega}$ (the closure of Ω). It is a complex since the range of the preceding map is contained in the kernel of the current map. Below we will work on other de Rham complexes where we keep the differential operators but change the spaces.

2.3. **Integration by Parts Formulas.** In order to prove commuting properties of projections we will need integration by parts. Let us state some useful integration by parts formulas.

$$\int_\Omega (\operatorname{div} F) v = -\int_\Omega F \cdot \operatorname{grad} v + \int_{\partial\Omega} (F \cdot n) v. \tag{2.1}$$

If F and G are vector valued functions we get

$$\int_\Omega \operatorname{curl} F \cdot G = \int_\Omega F \cdot \operatorname{curl} G + \int_{\partial\Omega} (n \times F) \cdot (n \times G \times n), \tag{2.2}$$

where n is the normal vector pointing out of $\partial\Omega$.

Now we state an integration by parts in a two dimensional domain. For simplicity we state it for a face f. We let curl_f be the surface curl and suppose we have that v is a vector valued function defined on f. Then,

$$\int_f (\operatorname{curl}_f v) = \int_{\partial f} v \cdot t, \tag{2.3}$$

where t unit tangent on ∂f.

Finally, we need the following identity. Let w be a vector valued function defined in $\mathbb{R}^3$ then

$$\operatorname{curl} w \cdot n = \operatorname{curl}_f (n \times w \times n) \qquad \text{on } f. \tag{2.4}$$

Hence,

$$\int_f \operatorname{curl} w \cdot n = \int_f \operatorname{curl}_f (n \times w \times n) = \int_{\partial f} (n \times w \times n) \cdot t = \int_{\partial f} w \cdot t \tag{2.5}$$

where n is the normal vector of our face f, and we used that $w \cdot t = (n \times w \times n) \cdot t$.

2.4. **Function Spaces.** The function spaces we will use are L^2 based. To L^2-norm on the domain Ω, is given by

$$\|u\|_{L^2(\Omega)} := \Big(\int_\Omega (u(x))^2 dx \Big)^{1/2},$$

where u is a scalar function.
We define the corresponding L^2 space as follows

$$L^2(\Omega) := \{u : \|u\|_{L^2(\Omega)} < \infty\}.$$

For a vector valued function we define

$$\|u\|_{L^2(\Omega)} := \Big(\int_\Omega |u(x)|^2 dx \Big)^{1/2},$$

where $|u(x)|^2 = u(x) \cdot u(x)$.

We now define three additional norms.

$$\|u\|^2_{H(\text{grad},\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\operatorname{grad} u\|^2_{L^2(\Omega)} \tag{2.6}$$

$$\|u\|^2_{H(\text{curl},\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\operatorname{curl} u\|^2_{L^2(\Omega)} \tag{2.7}$$

$$\|u\|^2_{H(\text{div},\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\operatorname{div} u\|^2_{L^2(\Omega)}. \tag{2.8}$$

We define the corresponding spaces as follows.

$$H(\text{grad},\Omega) := \{u : \|u\|_{H(\text{grad},\Omega)} < \infty\}, \tag{2.9}$$

$$H(\text{curl},\Omega) := \{u : \|u\|_{H(\text{curl},\Omega)} < \infty\}, \tag{2.10}$$

$$H(\text{div},\Omega) := \{u : \|u\|_{H(\text{div},\Omega)} < \infty\}. \tag{2.11}$$

Definition 2.4 (Inner Product in L^2)**.** The definition of the inner product for 2 functions in L^2 for an arbitrary domain Ω is given by

$$\langle u, v\rangle_\Omega = \int_\Omega u(x)v(x)\,dx$$

For vector valued functions, we need to use the dot product for the integrand.

Let us use the following notation.

$$W^0 = L^2(\Omega),\ W^1 = [L^2(\Omega)]^3,\ W^2 = [L^2(\Omega)]^3,\ W^3 = L^2(\Omega)$$

Where W^0 and W^3 are scalar function spaces and W^1 and W^2 are vector function spaces. We are going to use the following notation because it will be easier to state our results for the projections on what domains and output spaces we work with.

Next, we define the Sobolev spaces

$$V^0 = H(\text{grad},\Omega),\ V^1 = H(\text{curl},\Omega),\ V^2 = H(\text{div},\Omega),\ V^3 = L^2(\Omega),$$

where V^0 and V^3 are scalar function spaces and V^1 and V^2 are vector function spaces. We can think of these spaces as functions, where performing their respective operator, lie in their respective L^2 space. For example, in V^0, the gradient operator lies in W^1.

2.5. **p-chains and Boundary Operators.** We briefly recall p-chains and the boundary operators since our weight functions Z^k (see section 4) will have properties that need the boundary operator. Given the mesh $\mathcal{T}_h$ the space of p-chains defined on this mesh are the formal linear combination of p-subsimplices. To be more precise

$$C_p = \{\sum_{\sigma\in\Delta_p} a_\sigma\sigma : a_\sigma \in \mathbb{R}\}.$$

We note that Δ_p is a basis of C_p.

The boundary operator will be a linear operator that maps p-chains to $p-1$-chains. Since they are linear operator it is enough to define them on the basis Δ_p.

Definition 2.5 (Boundary Operator)**.** We have the boundary operator $\partial_p : C_p \to C_{p-1}$. We define on each basis element $\sigma \in \Delta_p$. Let $\sigma = [x_0, \dots, x_p]$ Then,

$$\partial_p\sigma := \sum_{i=0}^{p}(-1)^i[x_0, \dots, \widehat{x}_i, \dots, x_p]$$

where $\widehat{\cdot}$ denotes omission.

We introduce this well known theorem since it is analogous to the complex we are working with in this paper.

Theorem 2.6. *For all $c_p \in C_p$*

$$\partial_{p-1}\partial_p c_p = 0$$

We normally write the following to say that the space of chains form a complex:

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

3. Finite Element Exterior Calculus (FEEC) Spaces and Canonical Projections

3.1. **FEEC Spaces.** Now, we discuss the finite dimensional living inside our infinite spaces using the lowest-order finite element exterior calculus (FEEC) spaces in three dimensions. The second space is the lowest-order Nédélec finite element space (of the first kind) [11] and the third space is the Raviart-Thomas space. We will start by defining the local FEEC spaces along with the dergrees of freedom. The global FEEC spaces are defined by gluing together the local spaces using the degrees of freedom. Finally, we define the canonical interpolations I_h^k and a lemma that is represented as a diagram. These canonical interopolations gives us a preview of the projection Π_h^k, that we will be working with in this report.

Definition 3.1 (Local FEEC Spaces)**.** We define the corresponding spaces locally on each tetrahedra K. Note that $x = [x_1, x_2, x_3]$, where you treat the elements of x as variables.

$$\begin{aligned}
V_h^0(K) =& \{v : v(x) = a + b \cdot x, \text{ with } a \in \mathbb{R}, b \in \mathbb{R}^3\},\\
V_h^1(K) =& \{v : v(x) = a + b \times x, \text{ with } a \in \mathbb{R}^3, b \in \mathbb{R}^3\},\\
V_h^2(K) =& \{v : v(x) = a + bx \text{ with } a \in \mathbb{R}^3, b \in \mathbb{R}\},\\
V_h^3(K) =& \{v : v(x) = a \text{ with } a \in \mathbb{R}\}.
\end{aligned}$$

V_h^0 is the space of linear functions defined on the tetrahedron where a general element is of the form $v(x) = a_1 + b_1x_1 + b_2x_2 + b_3x_3$ for a $v \in V_h^k(K)$. Similarly $V_h^1(K)$ is known as the Nedelec space, and these are vector valued linear functions. The third space $V_h^2(K)$ is also vector valued and it is the well known Raviart-Thomas space [11]. Finally $V_h^3(K)$ are simply the constant functions.

Next, we define degrees of freedom (d.o.f) because they uniquely determine a function $V_h^i(K)$ where $i = 0, 1, 2, 3$ is the index. The degrees of freedom directly corresponds to the dimension of these spaces. These will also be useful to talk about continuity across our mesh.

Degrees of freedom for $V_h^0(K)$

$$\phi_y(v) = v(y), \text{ for all vertices } y \text{ of } K.$$

Since K has 4 vertices, the number of degrees of freedom are 4.

Degrees of freedom for $V_h^1(K)$

$$\phi_e(v) = \int_e v \cdot t, \text{ for all edges } e \text{ of } K. \tag{3.1}$$

Here t denotes a unit tangent vector to e. Since K has 6 edges, the number of degrees of freedom are 6.

Degrees of freedom for $V_h^2(K)$

$$\phi_f(v) = \int_f v \cdot n, \qquad \text{for all faces } f \text{ of } K. \tag{3.2}$$

Where n denotes a normal vector of the face f. Since K has 4 faces, the number of degrees of freedom are 4.

Degrees of freedom for $V_h^3(K)$

$$\phi_K(v) = \int_K v. \tag{3.3}$$

There is only 1 degree of freedom in this case.

We will have two equivalent ways of defining the Global FEEC spaces. We start with the first.

Definition 3.2 (Global FEEC Spaces)**.** We define our global FEEC spaces as follows:

$$\begin{aligned}
V_h^0 :=&\{v : v|_K \in V_h^0(K), \text{ for all } K \in \mathcal{T}_h, v \text{ is continuous across all vertices of } \mathcal{T}_h\},\\
V_h^1 :=&\{v : v|_K \in V_h^1(K), \text{ for all } K \in \mathcal{T}_h, v \times n \text{ is continuous across all faces of } \mathcal{T}_h\},\\
V_h^2 :=&\{v : v|_K \in V_h^2(K), \text{ for all } K \in \mathcal{T}_h, v \cdot n \text{ is continuous across all faces of } \mathcal{T}_h\},\\
V_h^3 :=&\{v : v|_K \in V_h^3(K), \text{ for all } K \in \mathcal{T}_h\}.
\end{aligned}$$

We now state an equivalent definition of (3.1) using degrees of freedom on the tetrahedra in our mesh. In particular we want to ensure continuity over multiple tetrahedra.

Definition 3.3 (Equivalent Definition of Global FEEC Spaces)**.**

$$V_h^k = \{v : v|_K \in V_h^k(K), \text{ for all } K \in \mathcal{T}_h, \text{ the corresponding d.o.f have the same value}\}$$

For example, when 2 tetrahedron meet (e.g T_1, T_2), 3 vertices meet at the face. If we have $v \in V_h^0$ then $v|_{T_1}$ and $v|_{T_2}$ must have the same values at the vertices. Similarly, we do the same for the edges with respect to the d.o.f. for V_h^1, then the faces for V_h^2.

3.2. **Canonical Interpolations.** We can now, using the degrees of freedom, define the canonical interpolations that can be used for the analysis. These link the infinite dimensional spaces to the finite dimensional spaces. It will be useful to define the following notation for these spaces as such:

$$S^0 := C^\infty(\overline{\Omega}),\ S^1 := [C^\infty(\overline{\Omega})]^3,\ S^2 := [C^\infty(\overline{\Omega})]^3,\ \text{ and } S^3 := C^\infty(\overline{\Omega}).$$

Definition 3.4 (I_h^k)**.** We define $I_h^k : S^k \to V_h^k$ as follows

$$I_h^0 v(y) = v(y) \qquad \text{for all } y \in \Delta_0, \tag{3.4a}$$

$$\int_e I_h^1 v \cdot t = \int_e v \cdot t \qquad \text{for all } e \in \Delta_1, \tag{3.4b}$$

$$\int_F I_h^2 v \cdot n = \int_F v \cdot n \qquad \text{for all } F \in \Delta_2, \tag{3.4c}$$

$$\int_K I_h^3 v = \int_K v \qquad \text{for all } K \in \Delta_3. \tag{3.4d}$$

Since $I_h^k v \in V_h^k$ and an a member of V_h^k is uniquely determined by the d.o.f defined above we see that we only need to specify the d.o.f. of $I_h^k v$ (which is what we do in the defintion). Moreover, from this we easily see that if $v \in V_h^k$ then $I_h^k v = v$.

Something useful from the way we define our I_h^k projections is the following fact about commuting.

Lemma 3.5. *It holds,*

$$\text{grad } I_h^0 v = I_h^1 \text{ grad } v \qquad \textit{for all } v \in S^0 \tag{3.5a}$$

$$\text{curl } I_h^1 v = I_h^2 \text{ curl } v \qquad \textit{for all } v \in S^1 \tag{3.5b}$$

$$\text{div } I_h^2 v = I_h^3 \text{div } v \qquad \textit{for all } v \in S^2. \tag{3.5c}$$

We omit the proof since this is not the interpolant we will be focused on in this paper. The proof can be found in Arnold, Falk, and Winther [4].

The lemma above is typically represented by a commuting diagram as follows

$$\begin{array}{ccccccc}
S^0 & \xrightarrow{\text{grad}} & S^1 & \xrightarrow{\text{curl}} & S^2 & \xrightarrow{\text{div}} & S^3\\
\downarrow{\scriptstyle I_h^0} & & \downarrow{\scriptstyle I_h^1} & & \downarrow{\scriptstyle I_h^2} & & \downarrow{\scriptstyle I_h^3}\\
V_h^0 & \xrightarrow{\text{grad}} & V_h^1 & \xrightarrow{\text{curl}} & V_h^2 & \xrightarrow{\text{div}} & V_h^3
\end{array}$$

Commutativity means that if one moves right then down on the diagram then this is equivalent to moving down then to the right. One of the drawbacks of I_h^k is that it not defined for all members of V^k.

4. Bounded Commuting Projections

Even though I_h^k satisfies commutativity and is a projection, the I_h^k interpolant is not bounded from V^k to V_h^k (in fact it might not even be defined). So, we now recall the projections Π_h^k [8]. These projections are commuting, and bounded in L^2. As mentioned in the introduction the existence of such projections are crucial for the analyis of FEM methods applied to various PDEs.

To be more precise, the projection in [8] satisfy the following.

The operators $\Pi_h^k : W^k \to V_h^k$, $k = 0, 1, 2, 3$ satisfy

$$\Pi_h^k v = v \qquad \text{if } v \in V_h^k \tag{4.1a}$$

$$\operatorname{grad} \Pi_h^0 v = \Pi_h^1 \operatorname{grad} v \qquad \text{for } v \in V^0 \tag{4.1b}$$

$$\operatorname{curl} \Pi_h^1 v = \Pi_h^2 \operatorname{curl} v \qquad \text{for } v \in V^1 \tag{4.1c}$$

$$\operatorname{div} \Pi_h^2 v = \Pi_h^3 \operatorname{div} v \qquad \text{for } v \in V^2 \tag{4.1d}$$

$$\|\Pi_h^k v\|_{L^2(\Omega} \leq C_k \|v\|_{L^2(\Omega)} \qquad \text{for } v \in W^k. \tag{4.1e}$$

Here (4.1a) is the projection property. That is, Π_h^k is the identity map when restricted to V_h^k. The equations (4.1b), (4.1c), and (4.1d) are the commuting properties. Finally, the inequality (4.1e) is the L^2 bounded property.

The key to the construction of the projections are weight functions Z^k. For each $\sigma \in \Delta_k$ $Z^k(\sigma)$ will be a scalar or vector valued weight function. More precisely:

$$Z^0(y) \in L^2(\Omega) \qquad \forall y \in \Delta_0, \tag{4.2}$$

$$Z^1(e) \in [L^2(\Omega)]^3 \qquad \forall e \in \Delta_1, \tag{4.3}$$

$$Z^2(F) \in [L^2(\Omega)]^3 \qquad \forall F \in \Delta_2, \tag{4.4}$$

$$Z^3(K) \in L^2(\Omega) \qquad \forall K \in \Delta_3. \tag{4.5}$$

Note that $Z^0(y)$ and $Z^3(K)$ are scalar valued functions and $Z^1(e)$ and $Z^2(F)$ are vector valued functions.

We define Z^k on k-chains by linearity on the simplices. For example, for Z^0 we have:

$$Z^0(\sum_y a_y y) = \sum_y a_y Z^0(y).$$

We mention this to show how we can combine 0-chains on Z^0. Similarly we can define Z^k on k-chains.

The projections Π^k then take the following form.

Definition 4.1 (Π_h^k Projections)**.** The Π_h^k projections take the following form:

$$\Pi_h^0 v(y) = \int_\Omega (Z^0(y))(x) v(x) dx \qquad \forall y \in \Delta_0, \tag{4.6}$$

$$\int_e \Pi_h^1 v \cdot t_e = \int_\Omega (Z^1(e))(x) \cdot v(x) dx \qquad \forall e \in \Delta_1, \tag{4.7}$$

$$\int_F \Pi_h^2 v \cdot n_F = \int_\Omega (Z^2(F))(x) \cdot v(x) dx \qquad \forall F \in \Delta_2, \tag{4.8}$$

$$\int_K \Pi_h^3 v = \int_\Omega (Z^3(K))(x) v(x) dx \qquad \forall K \in \Delta_3. \tag{4.9}$$

The projections are defined on the d.o.f, but this time we take an integral over Ω, and this allowed one to act on L^2 functions. This will allow functions that even have jump discontinuities.

If we want to see the projection written more explicitly instead of just stating their d.o.f we need the so-called Whitney forms.

Definition 4.2 (Whitney Forms)**.** These functions serve as a basis for our FEEC spaces as the following:

$$W^0(y) \in V_h^0 \quad (W^0(y))(z) = \begin{cases} 1 & \text{if } z = y \\ 0 & \text{if } z \neq y \end{cases} \tag{4.10}$$

$$W^1(e) \in V_h^1 \quad \int_g W^1(e) \cdot t_g = \begin{cases} 1 & \text{if } g = e \\ 0 & \text{if } g \neq e \end{cases} \tag{4.11}$$

$$W^2(F) \in V_h^2 \quad \int_H (W^2(F)) \cdot n_H = \begin{cases} 1 & \text{if } H = F \\ 0 & \text{if } H \neq F \end{cases} \tag{4.12}$$

$$W^3(K) \in V_h^3 \quad \int_L (W^3(K)) = \begin{cases} 1 & \text{if } L = K \\ 0 & \text{if } L \neq K. \end{cases} \tag{4.13}$$

Where g is any edge, H is any face, and L is any tetrahedron on our mesh.

We then have Π_h^k written as:

$$\begin{aligned} \Pi_h^0 v &= \sum_{y \in \Delta_0} \langle Z^0(y), v \rangle W^0(y) \\ \Pi_h^1 v &= \sum_{e \in \Delta_1} \langle Z^1(e), v \rangle W^1(e) \\ \Pi_h^2 v &= \sum_{F \in \Delta_2} \langle Z^2(F), v \rangle W^2(F) \\ \Pi_h^3 v &= \sum_{K \in \Delta_3} \langle Z^3(K), v \rangle W^3(K). \end{aligned}$$

The next lemma shows what is needed from the weights Z^k to ensure that the projection Π^k are commuting.

Lemma 4.3. *If the weights satisfy the following properties:*

$$-\text{div } Z^1(e) = Z^0(\partial_1 e), \qquad \textit{for all } e \in \Delta_1, \tag{4.14}$$

$$\text{curl } Z^2(F) = Z^1(\partial_2 F), \qquad \textit{for all } F \in \Delta_2, \tag{4.15}$$

$$-\text{grad } Z^3(K) = Z^2(\partial_3 K) \qquad \textit{for all } K \in \Delta_3, \tag{4.16}$$

then, the projections satisfy the commuting properties (4.1b),(4.1c)*, and* (4.1d)*.*

Proof. We only prove (4.1d) as the proof of the other identities are similar. We need to show

$$\text{div } \Pi_h^2 v = \Pi_h^3 \text{div } v \qquad \text{if } v \in V_h^2.$$

provided

$$-\text{grad } Z^3(K) = Z^2(\partial_3 K) \qquad \text{for all } K \in \Delta_3.$$

Since $\text{div } \Pi_h^2 v$ and $\Pi_h^3 \text{div } v$ belong to V_h^3, it is enough to show that the d.o.f match:

$$\int_K \text{div } \Pi_h^2 v = \int_K \Pi_h^3 \text{div } v \qquad \text{for all} K \in \Delta_3. \tag{4.17}$$

To this end, let $\partial_3 K = F_1 + F_2 + F_3 + F_4$ then the left hand side can be written as

$$\begin{aligned} \int_K \text{div } \Pi_h^2 v &= \int_{\partial K} \Pi_h^2 v \cdot n \\ &= \int_{\sum_{i=1}^4 F_i} \Pi_h^2 v \cdot n \\ &= \sum_{i=1}^4 \int_{F_i} \Pi_h^2 v \cdot n. \end{aligned}$$

In the first step we used integration by parts (2.1).

If we use the definition of Π_h^2 we have

$$\int_{F_i} \Pi_h^2 v \cdot n = \int_\Omega (Z^2(F_i))(x) \cdot v(x)\, dx.$$

Hence, we have

$$\begin{aligned}\sum_{i=1}^4 \int_{F_i} \Pi_h^2 v \cdot n &= \sum_{i=1}^4 \int_\Omega Z^2(F_i)(x) \cdot v(x)\, dx \\ &= \int_\Omega Z^2(\sum_{i=1}^4 F_i)(x) \cdot v(x)\, dx \\ &= \int_\Omega Z^2(\partial K)(x) \cdot v(x)\, dx \\ &= -\int_\Omega \operatorname{grad} Z^3(K)(x) \cdot v(x)\, dx.\end{aligned}$$

In the last step we used (4.16). Using integration by parts (2.1) again we have

$$-\int_\Omega (\operatorname{grad} Z^3(K)(x) \cdot v(x)\, dx = \int_\Omega Z^3(K) \operatorname{div} v(x)\, dx,$$

where we used that $Z^3(K)$ vanishes on $\partial\Omega$.

Thus, we have showed

$$\int_K \operatorname{div}\, \Pi_h^2 v = \int_\Omega Z^3(K) \operatorname{div} v(x)\, dx.$$

Using the definition of Π^3 gives

$$\int_\Omega Z^3(K) \operatorname{div} v(x)\, dx = \int_\Omega \Pi_h^3 \operatorname{div} v(x)\, dx,$$

which gives (4.17).

□

Now let us investigate what what properties on the weight functions do we need if we would like Π_h^k to satisfy the projection property (i.e. (4.1a)).

Lemma 4.4. *Suppose that*

$$\begin{aligned}\int_\Omega (Z^0(y))(x) v(x) dx &= v(y) && \forall v \in V_h^0, \forall y \in \Delta_0(\mathcal{T}_h), \\ \int_\Omega (Z^1(e))(x) \cdot v(x) dx &= \int_e v \cdot t_e && \forall v \in V_h^1, \forall e \in \Delta_1(\mathcal{T}_h), \\ \int_\Omega (Z^2(F))(x) \cdot v(x) dx &= \int_F v \cdot n_F && \forall v \in V_h^2, \forall F \in \Delta_2(\mathcal{T}_h), \\ \int_\Omega (Z^3(K))(x) v(x) dx &= \int_K v && \forall v \in V_h^3, \forall K \in \Delta_3(\mathcal{T}_h).\end{aligned}$$

Proof. Let $v \in V_h^k$. We notice that our assumptions match exactly to the degrees of freedom of their respected V_h^k space hence we can see by that since the matching degrees of freedom from $\Pi_h^k v$ match exactly the same degrees of freedom that uniquely determine $v \in V_h^k$ so it maps v directly to itself. □

Conditions on the weights to guarantee the bounded property (4.1e) is somewhat more involved and do not include here but instead refer to the paper [8].

5. Construction of the weights

To construct our weights, we first need to define star of vertex y and extended star of $\sigma \in \Delta_\ell$ as follows.

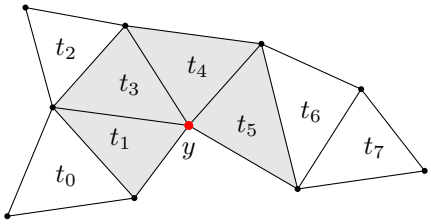
Definition 5.1 (Star of y and extended star of σ)**.**

$$\mathrm{st}(y) = \bigcup_{\substack{\tau \in \mathcal{T}_h \\ y \in \Delta_0(\tau)}} \tau, \quad \mathrm{es}(\sigma) = \bigcup_{\substack{y \in \Delta_0(\sigma) \\ \sigma \cap \tau \neq \emptyset}} \mathrm{st}(y).$$

We can think of the $\mathrm{st}(y)$ as the region in $\mathcal{T}_h$ that intersects the vertex y. The extended star of y is the region that intersects with the vertices of σ. For example, if $\sigma \in \Delta_1$ and $\sigma = [y_1, y_2]$, then $\mathrm{es}(\sigma)$ is the region that intersects with both y_1 and y_2.

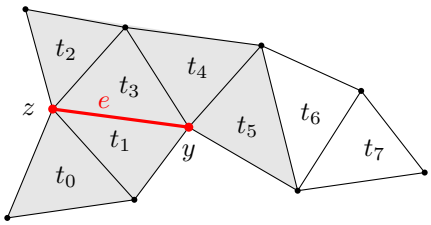
To illustrate these concepts we provide some examples.

Example 5.2. For $y \in \Delta_0$ in a 2D mesh the $\mathrm{st}(y)$ is illustrated as follows:



When looking at the vertex $y \in \Delta_0$ the $\mathrm{st}(y)$ is the region t_1, t_3, t_4, t_5. Note that $st(y) = es(y)$ in this example.

Example 5.3. We will now examine $es(e)$ in a 2D mesh where $e \in \Delta_1$ and $e = [y, z]$ in the diagram below.



When looking at the edge $e \in \Delta_1$ the $es(e)$ is the region $t_0, t_1, t_2, t_3, t_4, t_5$. The $st(e) \neq es(e)$ in this example.

We need to consider the space of V_h^k restricted to extend stars.

Definition 5.4 ($V_h^k(es(\sigma))$)**.** $V_h^k(\mathrm{es}(\sigma)) := \{v|_{\mathrm{es}(\sigma)} : v \in V_h^k\}$

$V_h^k(\mathrm{es}(\sigma))$ is the set of all functions obtained by taking a function from V_h^k and looking only at its values on the extended star of Ω.

We will need the kernel of these spaces.

Definition 5.5 (Kernels of the local spaces)**.**

$$\mathfrak{Z}V_h^0(\mathrm{es}(y)) = \{v \in V_h^0(\mathrm{es}(y)) : \mathrm{grad}\, v = 0\},$$
$$\mathfrak{Z}V_h^1(\mathrm{es}(e)) = \{v \in V_h^1(\mathrm{es}(e)) : \mathrm{curl}\, v = 0\},$$
$$\mathfrak{Z}V_h^2(\mathrm{es}(F)) = \{v \in V_h^2(\mathrm{es}(F)) : \mathrm{div}\, v = 0\}.$$

Note that $\mathfrak{Z}V_h^0(\mathrm{es}(y)) = \{v \in V_h^0(\mathrm{es}(y)) : \mathrm{grad}\, v = 0\} = \ker(\mathrm{grad}, V_h^0(\mathrm{es}(y))$. Since only the constant functions belong to the kernel of the gradient we see that $\mathfrak{Z}V_h^0(\mathrm{es}(y)) = \mathbb{R}$.

We also need the orthogonal compliment of this space.

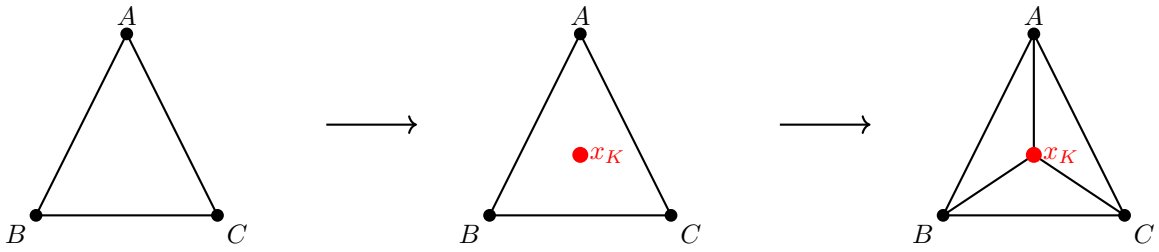
Definition 5.6 (Orthogonal Compliment of the Kernel)**.**

$$\mathfrak{Z}^\perp V_h^0(\mathrm{es}(y)) = \{v \in V_h^0(\mathrm{es}(y)) : \langle v, u\rangle_{\mathrm{es}(y)} = 0, \forall u \in \mathfrak{Z}V_h^0 \mathrm{es}(y)\},$$
$$\mathfrak{Z}^\perp V_h^1(\mathrm{es}(e)) = \{v \in V_h^1(\mathrm{es}(e)) : \langle v, u\rangle_{\mathrm{es}(e)} = 0, \forall u \in \mathfrak{Z}V_h^1(\mathrm{es}(e))\},$$
$$\mathfrak{Z}^\perp V_h^2(\mathrm{es}(F)) = \{v \in V_h^2(\mathrm{es}(F)) : \langle v, u\rangle_{\mathrm{es}(F)} = 0, \forall u \in \mathfrak{Z}V_h^2(\mathrm{es}(F))\}.$$

The space $\mathfrak{Z}^{\perp}V_h^0(\mathrm{es}(y))$ is called the *orthogonal complement* of the kernel. It consists of all functions in $V_h^0(\mathrm{es}(y))$ that are orthogonal to every function in $\mathfrak{Z}V_h^0$ with respect to the chosen L^2-inner product.
For example, since $\mathfrak{Z}V_h^0(\mathrm{es}(y)) = \mathbb{R}$ we have that

$$\mathfrak{Z}^{\perp}V_h^0(\mathrm{es}(y)) = \{v \in V_h^0(\mathrm{es}(y)) : \int_{\mathrm{es}(y)} v(x)dx = 0\}.$$

We will also need the FEEC spaces on a refinement of our mesh $\mathcal{T}_h$. From our original mesh $\mathcal{T}_h$ we define the Alfeld mesh $\mathcal{T}_h^A$ by choosing the barycenter, denoted x_K, for each $K \in \mathcal{T}_h$ connecting to all the vertices of K in a straight line. The corresponding FEEC spaces on the new mesh $\mathcal{T}_h^A$ will be denoted by $V_{h,A}^k$.



Original element K Choose barycenter Alfeld split of K

FIGURE 1. Construction of the Alfeld split. Starting from a simplex K, its barycenter x_K is introduced and connected to each vertex, subdividing K into smaller simplices. Repeating this process for every element of the mesh $\mathcal{T}_h$ produces the Alfeld mesh $\mathcal{T}_h^A$.

We will also need the restrictions to extend stars with zero boundary conditions.

Definition 5.7 (Restrictions of Local spaces on Alfeld split)**.** We define:

$$\mathring{V}_{h,A}^0(\mathrm{es}(y)) = \{v \in V_{h,A}^2(\mathrm{es}(y)) : v = 0,\ \partial\mathrm{es}(y)\} \tag{5.1}$$

$$\mathring{V}_{h,A}^1(\mathrm{es}(e)) = \{v \in V_{h,A}^2(\mathrm{es}(e)) : v \times n = 0,\ \partial\mathrm{es}(e)\} \tag{5.2}$$

$$\mathring{V}_{h,A}^2(\mathrm{es}(F)) = \{v \in V_{h,A}^2(\mathrm{es}(F)) : v \cdot n = 0,\ \partial\mathrm{es}(F)\} \tag{5.3}$$

$$\mathring{V}_{h,A}^3(\mathrm{es}(K)) = \{v \in V_{h,A}^3(\mathrm{es}(K)) : \int_{\mathrm{es}(K)} v = 0\} \tag{5.4}$$

We are now ready to define the weight functions. We start with Z^0.

5.1. **Construction of $Z^0(y)$.** Our goal is to construct $Z^0(y)$ for any $y \in \Delta_0$. By obtaining our Z^0 weight, we can find the Z^1, Z^2 and Z^3 weights.
The first weight Z^0 is defined by this formula:

$$Z^0(y) = \eta_0(y) - \mathrm{div}\,(\mu_y \,\mathrm{grad}\,\psi_0(y))$$

To use this formula we need to introduce $\eta_0(y)$, μ_y and $\psi_0(y)$.

We introduce the indicator function now, for the definitions that follow shortly after it.

Definition 5.8 ($\chi_A(x)$)**.** Definition of indicator function on general set A

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

We also need the piecewise linear functions defined on the Alfeld split that vanish on the boundary of each $K \in \Delta_3(\mathcal{T}_h)$ and takes values on each barycenter x_K.

Definition 5.9 (μ)**.** Let $\mu \in V_{h,A}^0$ be such that μ vanishes at all the vertices of the original mesh $\mathcal{T}_h$ and takes value one on the barycenter x_K for all $K \in \mathcal{T}_h$.

For a given vertex $y \in \Delta_0$ we define μ_y which is μ restricted to $\mathrm{es}(y)$ and then extend to zero outside of $\mathrm{es}(y)$.

Definition 5.10 (μ_y)**.** For a vertex y of $\mathcal{T}_h$ we let

$$\mu_y = \mu\chi_{\mathrm{es}(y)}.$$

The function $\eta_0(y)$ will be piece-wise constant function defined on $\mathcal{T}_h^A$.

Definition 5.11 ($\eta_0(y)$)**.** For $y \in \Delta_0$ we define

$$\eta_0(y) = \frac{1}{|\mathrm{es}(y)|}\chi_{\mathrm{es}(y)}$$

where $|\mathrm{es}(y)| = \int_{\mathrm{es}(y)} 1dx$ indicates the volume of $\mathrm{es}(y)$. Thus,

$$(\eta_0(y))(x) = \begin{cases} \frac{1}{|\mathrm{es}(y)|} & \text{if } x \in \mathrm{es}(y) \\ 0 & \text{if } x \notin \mathrm{es}(y). \end{cases}$$

Hence, we see that $\eta_0(y)$ is simply a piece-wise constant function on the mesh $\mathcal{T}_h^A$ and $\int_\Omega \eta_0(y) = 1$. Therefore, $\eta_0(y)$ is very simple to construct numerically.

It remains to specify $\psi_0(y)$. It is not given explicitly, it has to be solved using a linear system:

The function $\psi_0(y) \in \mathfrak{Z}^\perp V_h^0(\mathrm{es}(y))$ satisfies

$$\langle \mu_y \operatorname{grad} \psi_0(y), \operatorname{grad} u\rangle_{\mathrm{es}(y)} = u(y) - \langle \eta_0(y), u\rangle_{\mathrm{es}(y)} \qquad \forall u \in \mathfrak{Z}^\perp V_h^0(\mathrm{es}(y)).$$

This is also not so difficult to construct numerically since $\mathfrak{Z}^\perp V_h^0(\mathrm{es}(y))$ are simply members of $V_h^0(\mathrm{es}(y))$ that are orthogonal to constant functions.

5.2. **Construction of $Z^1(e)$.** Now we move on to our second weight Z^1 which takes the form:

$$Z^1(e) = \eta_1(e) + \operatorname{curl}(\mu_e \operatorname{curl} \psi_1(e)).$$

Definition 5.12 (μ_e)**.** For an edge e of $\mathcal{T}_h$ we let

$$\mu_e = \mu\chi_{\mathrm{es}(e)}.$$

where we recall that $\chi_{\mathrm{es}(e)}$ is the indicator function of $\mathrm{es}(e)$.

The Raviart-Thomas element $\eta_1(e) \in \mathring{V}^2_{h,A}$ will satisfy the following properties:

$$-\operatorname{div} \eta_1(e) = Z^0(\partial_1 e) \qquad \text{in } \mathrm{es}(e) \tag{5.5a}$$

$$\eta_1(e) \cdot n = 0, \qquad \text{on } \partial\mathrm{es}(e) \tag{5.5b}$$

$$\eta_1(e) = 0, \qquad \text{in } \Omega \setminus \mathrm{es}(e) \tag{5.5c}$$

$$\|\eta_1(e)\| \le C h_e \|Z^0(\partial e)\|, \tag{5.5d}$$

where n denotes the normal vector on the boundary $\partial\,\mathrm{es}(e)$, and $h_e = |e|$ is the length of the edge e.

A way we propose to solve for such vector field is to solve a mixed problem:

Find $\eta_1(e) \in \mathring{V}^2_{h,A}(es(e))$, $u \in \mathring{V}^3_{h,A}(\mathrm{es}(e))$ such that:

$$\langle \eta_1(e), v\rangle_{\mathrm{es}(e)} + \langle u, \operatorname{div} v\rangle_{\mathrm{es}(e)} = 0 \qquad \forall v \in \mathring{V}^2_{h,A}(\mathrm{es}(e)), \tag{5.6a}$$

$$-\langle \operatorname{div} \eta_1(e), w\rangle_{\mathrm{es}(e)} = \langle Z^0(\partial e), w\rangle \qquad \forall w \in \mathring{V}^3_{h,A}(\mathrm{es}(e)). \tag{5.6b}$$

Ideally, we would only need to solve only the second equation (5.6b). However, this will lead to a non-square system that might not be solvable. Hence, we need to augment the system and add a new unknown u which gives a square system (5.6). Although we solve for both $\eta_1(e)$ and u we only use $\eta_1(e)$ and discard u. We extend $\eta_1(e)$ outside of $\mathrm{es}(e)$ by zero (hence, (5.5c) is satisfied). By definition (5.5b) is satisfied. We see that (5.5a) holds since we can choose $w = -\operatorname{div} \eta_1(e) - Z^0(\partial e)$ in (5.6b) and get

$$\langle w, w\rangle_{\mathrm{es}(e)} = \int_{es(e)} w^2 = 0,$$

which implies $w = 0$ and in turn shows (5.5a). The estimate (5.5d) is more delicate and we omit the details.

From a computational standpoint we solve for this small linear system for every edge e to obtain $\eta_1(e)$.

It remains to specify $\psi_1(e)$. It satisfies the following linear system:

Find $\psi_1(e) \in \mathfrak{Z}^\perp V_h^1(\mathrm{es}(e))$ satisfying

$$\langle \mu_e \operatorname{curl} \psi_1(e), \operatorname{curl} u\rangle_{\mathrm{es}(e)} = \phi_e(u) - \langle \eta_1(e), u\rangle_{\mathrm{es}(e)} \qquad \forall u \in \mathfrak{Z}^\perp V_h^1(\mathrm{es}(e)). \tag{5.7}$$

This leads to a square linear system. The difficulty arises since it is not straightforward to obtain a basis for the orthogonal complement space $\mathfrak{Z}^\perp V_h^1(\mathrm{es}(e))$.

5.3. **Construction of $Z^2(F)$.** Using the same idea as before. Our $Z^2(F) \in \mathring{V}_{h,A}^1(\mathrm{es}(F))$ weight for a $F \in \Delta_2$ takes the following form:

$$Z^2(F) = \eta_2(F) - \operatorname{grad}(\mu_F \operatorname{div} \psi_2(F)). \tag{5.8}$$

Note that the boundary conditions $Z^2(F) \times n = 0$ on $\partial \mathrm{es}(F)$ are enforced.

We first need to find $\eta_2(F) \in \mathring{V}_{h,A}^1(\mathrm{es}(F))$ such that:

$$\operatorname{curl} \eta_2(F) = Z^1(\partial F).$$

As proposed by Boffi [6], for example, from a computational point of view we propose to solve the following mixed problem:

Find $\eta_2(F) \in \mathring{V}_{h,A}^1(es(F))$, $u \in \mathring{V}_{h,A}^2(\mathrm{es}(F))$ such that:

$$\langle \eta_2(F), v\rangle_{\mathrm{es}(F)} + \langle u, \operatorname{curl} v\rangle_{\mathrm{es}(F)} = 0 \qquad \forall v \in \mathring{V}_{h,A}^1(\mathrm{es}(F)), \tag{5.9a}$$

$$\langle \operatorname{curl} \eta_2(F), w\rangle_{\mathrm{es}(F)} = \langle Z^1(\partial_2 F), w\rangle \qquad \forall w \in \mathring{V}_{h,A}^2(\mathrm{es}(F)). \tag{5.9b}$$

Again, as we did with $\eta_1(e)$ we needed to augment the system to get a square system. We only use $\eta_2(F)$ and discard u.

It remains to construct $\psi_2(F)$.

The function $\psi_2(F) \in \mathfrak{Z}^\perp V_h^2(\mathrm{es}(F))$ satisfies

$$\langle \mu_F \operatorname{div} \psi_2(F), \operatorname{div} u\rangle_{\mathrm{es}(F)} = \phi_F(u) - \langle \eta_2(F), u\rangle_{\mathrm{es}(F)} \qquad \forall u \in \mathfrak{Z}^\perp V_h^2(\mathrm{es}(e)). \tag{5.10}$$

Again, computational the difficulty is finding a basis for $\mathfrak{Z}^\perp V_h^2(\mathrm{es}(F))$.

5.4. **Construction of $Z^3(K)$.** Finally, the weight $Z^3(K)$ takes the form:

$$Z^3(K) = \eta_3(K),$$

where $\eta_3(K) \in \mathring{V}_{h,A}^0(es(K))$:

$$-\operatorname{grad} \eta_3(K) = Z^2(\partial_3 K) \quad \text{ on } \mathrm{es}(K).$$

In this case, computational we propose to solve:

Find $\eta_3(K) \in \mathring{V}_{h,A}^0(\mathrm{es}(K))$, such that:

$$\langle \operatorname{grad} \eta_3(K), \operatorname{grad} w\rangle_{\mathrm{es}(K)} = -\langle Z^2(\partial_3 K), \operatorname{grad} w\rangle_{\mathrm{es}(K)} \qquad \forall w \in \mathring{V}_{h,A}^0(\mathrm{es}(K)).$$

This leads to a square, positive definite system. Note that since $\operatorname{curl} Z^2(\partial_3 K) = Z^1(\partial_2 \partial_3 K) = Z^1(0) = 0$ we have that $Z^2(\partial_3 K) = \operatorname{grad} m$ for some $m \in \mathring{V}_{h,A}^0(\mathrm{es}(K))$. Hence, this implies that indeed $-\operatorname{grad} \eta_3(K) = Z^2(\partial_3 K)$.

6. Implementation

As we have seen above in order to construct the projections Π_h^k we need to build the weights Z^k. In the previous sections we discussed the construction of the weights Z^k. So far, we have implemented Z^0 and Z^1 using NGSolve [12] using the ideas discussed above. We hope to implement the remaining projections (i.e. Π_h^2, Π_h^3) in the near future. You can find our code in Github.

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